

MTU_ValueService **Technical Documentation**

Electronic Equipment
Serial Interface

CANopen, J1939, SAM Advanced
Application: Genset

Functional Description
E532337/01E



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1	CANopen.....	5
1.1	Transmission values – CANopen Genset.....	5
1.2	Binary values — Transmitting (ECU/SAM to external device/system)	12
1.3	Analog values — Transmitting (ECU/SAM to external device/system)	14
1.4	Diagnostic messages — Transmitting (ECU/SAM to an external device/system)	52
1.5	Binary setpoint signals/values — Receive (external device/system to ECU/SAM)	58
1.6	Analog setpoint signals/values — Receive (external device/system to ECU/SAM)	60
2	J1939.....	67
2.1	Transmission values J1939.....	67
2.2	J1939 Standard — Transmitting (MTU to external devices).....	69
2.3	J1939 Standard — Receiving (external device to MTU).....	71
2.4	Proprietary objects — Transmitting (MTU to external devices)	72
2.5	Proprietary objects — Receiving (external device to MTU)	80
3	Appendix A.....	83
3.1	Abbreviations	83
3.2	Conversion tables.....	87
3.3	MTU contact persons/service partners.....	91
4	Appendix B.....	93
4.1	Index.....	93

1 CANopen

1.1 Transmission values – CANopen Genset

General information about the interface

Purpose

MTU provides an interface for genset applications in accordance with the CANopen protocol.

This document sets out the Process Data Objects (abbreviated in the following as PDOs), which are transmitted in accordance with this protocol.

The PDOs, which are up to 8 bytes in length, with a total of 64 bits, can transmit the following values:

- Binary values
- Analog values
- Diagnostic messages
- Binary setpoints/values
- Analog setpoints/values

Physical specifications

- CAN specification: ISO11898, CAN 2.0A/B, electrically isolated, 3-wire
- The following node and baud rate settings apply:
 - Node no.: 6
 - Baud rate: 125Kb
- 11 bit identifier
- Start/Stop CANopen devices

CANopen NMT and communication

- CANopen specification: DS301 V 4.01
- The MTU system operates as a slave and waits to be started by a master.
- Static mapping and dynamic mapping are not supported.
- SDOs are not supported
- No write access to the object directory via SDOs
- Support of standard PDOs
- Transmitted data format: Intel format
- The NMT monitoring takes place exclusively by means of a heartbeat, Heartbeat Producer Time: 500 ms
- Transmission Type 255
 - Inhibit time: 250 ms
 - Event time 20 s
- Analog values are transmitted in Unsigned32 format
- Boolean values are transmitted in Boolean format
- IMPORTANT: MTU uses data packages with less than 8 data bytes. The statement is held in the DCF/EDS file compiled by MTU (provided) (→ see also PDO tables).

Missing data detection, Missing Data (MD)/Sensor Defect (SD)

The following values are transmitted in case of faults:

- Analog values:
 - MD value: 0x7FFFFFFF
 - SD value: 0x80000001
- Boolean values

The standard dictates that in the case of PDOs which transmit binary values, bytes 0-3 contain the user data and 4-7 the status values:

The table below shows the PDO structure for Boolean values.

PDO structure for Boolean values							
Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8

PDO structure for Boolean values							
MV	MV	MV	MV	MD/SD status	MD/SD status	MD/SD status	MD/SD status

Table 1: PDO structure for Boolean values

Example:

Measurand in byte 1 is e.g. 0

The associated status is given in byte 5:

- e.g. 0 for measurand is OK
- e.g. 1 for measurand is MD/SD

Notes

Notes on the tables

In the headlines of the tables * appear at various positions. These have the following significance:

- * One of the following fault codes: 134, 136, 142, 144, 145, 147, 149, 151, 363, 364
- ** Takes the value ZERO or 1
 - additionally, with Sensor Defect (SD) → ZERO
 - in the case of Missing Data (MD) → 1
- *** Prerequisite for this COB ID is: NODE ID = 11. For other Node ID (→ see calculation of the COB IDs)
- **** ERR_MD is indicated if PV110010 MD is → ERR_MD = 1. Combination ERR_MD = 1 and PV510037 bit 0 = 1 → Communication error

Calculation of the COB IDs

The CAN_IDs must be calculated as follows:

TXPDO(n): tx_Base_identifier/s + Offset + Node_ID

RXPDO(n): rx_Base_identifier/s + Offset + Node_ID

for n= 1.. 4:

- tx_Base_identifier (1..4) = 0x180, 0x280, 0x380, 0x480
- rx_Base_identifier (1..4) = 0x200, 0x300, 0x400, 0x500

Transmit PDOs

Table columns PDO Binary values — Transmitting

The table heading for the binary messages (transmitting) has the following structure:

Name	PV No.	Byte	Byte Bit No.	PD Bit No.	Obj. Bit No.	PDO-Bin. Status message: NODE ID + 0, Transmit PDO 1; COB ID 186***	PV		Comment
							Sys. No.	Index	

Table 2: Table columns PDO Binary values — Transmitting

The columns have the following meaning:

- Name
Name of the process variables
- PV No.
Number of the process variables
- Byte
Number of the byte
- Byte Bit No.
Number of the bit within the corresponding byte
- PD Bit No.
Number of the bit, referenced to the complete PDO
- Object Bit No.
Number of the bit, referenced to the object
- PDO type and COB ID
e.g.:
PDO binary status signal: describing the type of process data object
NODE ID + 0, Transmit PDO 1; : Statement of the value
COB ID 186***: Statement of the value and further information on ***
The PV is described in detail in the field below
- PV, this number comprises:
 - System number (MTU internal number in the engine governor)
 - System Index (MTU internal number in the engine governor)
- Comment
If required, further information can be given here

Table Columns PDO analog values — Transmitting

The table heading for analog values (transmit) has the following structure:

Name	PV No.	Byte	Byte Bit No.	PD Bit No.	Obj. Bit No.	PDO analog measured values: NODE ID + 0, Transmit PDO 2; COB ID 286***	Comment
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Table 3: Table Columns PDO analog values — Transmitting

The columns have the following meaning:

- Name
Name of the process variables
- PV No.
Number of the process variables
- Byte
Number of the byte
- Byte Bit No.
Number of the bit within the corresponding byte
- PD Bit No.
Number of the bit, referenced to the complete PDO
- Object Bit No.
Number of the bit, referenced to the object
- PDO type and COB ID
e.g.:
PDO analog measured values: describing the type of process data object
NODE ID + 0, Transmit PDO 2; : Statement of the value
COB ID 286***: Statement of the value and further information on ***
The PV is described in detail in the field below
- Comment
If required, further information can be given here

Diagnostic Messages — Transmit

The table heading for binary diagnostic messages (transmit) has the following structure:

Name	PV No.	Byte	Byte Bit No.	PD Bit No.	Obj. Bit No.	PDO-Bin. Diagnostic message: NODE ID + 5, Transmit PDO 1; COB ID 18B***	PV		Comment
							Sys. Index	PV Index	

Table 4: Diagnostic Messages — Transmitting

The columns have the following meaning:

- Name
Name of the process variables
- PV No.
Number of the process variables
- Byte
Number of the byte
- Byte Bit No.
Number of the bit within the corresponding byte
- PD Bit No.
Number of the bit, referenced to the complete PDO
- Object Bit No.
Number of the bit, referenced to the object
- PDO type and COB ID
e.g.:
PDO-Bin. Diagnostic message: describing the type of process data object
NODE ID + 5, Transmit PDO 1; COB ID 18B*** statement of the values and further information on ***
The PV is described in detail in the field below
- PV, this number comprises:
 - System number (MTU internal number in the engine governor)
 - System Index (MTU internal number in the engine governor)
- Comment
If required, further information can be given here

Binary setpoint signals/values — Receive

The table heading for binary data (Received) has the following structure:

Name	PV No.	Byte	Byte Bit No.	PD Bit No.	Obj. Bit No.	PDO-Bin. Spec.: NODE ID + 0, Receive PDO 1; COB ID 206***	PV		Comment
							Sys. Index	PV Index	

Table 5: Binary setpoint signals/values — Receiving

The columns have the following meaning:

- Name
Name of the process variables
- PV No.
Number of the process variables
- Byte
Number of the byte
- Byte Bit No.
Number of the bit within the corresponding byte
- PD Bit No.
Number of the bit, referenced to the complete PDO
- Object Bit No.
Number of the bit, referenced to the object
- PDO type and COB ID
e.g.:
PDO-Bin. Spec.: describing the type of process data object
NODE ID + 0, Receive PDO 1; COB ID 206***: Statement of the values and further information on ***
The PV is described in detail in the field below
- PV, this number comprises:
 - System number (MTU internal number in the engine governor)
 - System Index (MTU internal number in the engine governor)
- Comment
If required, further information can be given here

Analog setpoint signals/values — Receive

The table heading for analog values (Received) has the following structure:

Name	PV No.	Byte	Byte Bit No.	PD Bit No.	Obj. Bit No.	PDO analog setpoint signals: NODE ID + 0, Receive PDO 2; COB ID 306***	Comment
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Table 6: Analog setpoint signals/values — Receiving

The columns have the following meaning:

- Name
Name of the process variables
- PV No.
Number of the process variables
- Byte
Number of the byte
- Byte Bit No.
Number of the bit within the corresponding byte
- PD Bit No.
Number of the bit, referenced to the complete PDO
- Object Bit No.
Number of the bit, referenced to the object
- PDO type and COB ID
e.g.:
PDO analog setpoint signals: describing the type of process data object
NODE ID + 0, Receive PDO 2; : Statement of the value
COB ID 306***: Statement of the value and further information on ***
The PV is described in detail in the field below
- Comment
If required, further information can be given here

Overview

The following tables provide an overview of all transmit and receive PDOs. Each PDO can be transmitted up to two process variables (PVs).

Transmit PDOs (Transmit)

Node ID Offset 0

Node ID Offset	Transmit PDO	COB ID	PV Name	PV Name
0	1	186	PV500030 Binary Signal 3	PV500031 Binary Status3
0	2	286	PV001140 T-Lube Oil	PV001126 T-Coolant
0	3	386	PV001002 Engine Speed	PV001006 Nominal Speed
0	4	486	PV001115 ECU Operating Hours	PV001005 Idle Speed

Table 7: Node ID Offset 0

Node ID Offset 1

Node ID Offset	Transmit PDO	COB ID	PV Name	PV Name
1	1	187	PV001026 P-Lube Oil	PV001049 P-Charge Air
1	2	287	PV001131 T-Charge Air	PV001152 T-Fuel
1	3	387	PV001046 P-Fuel	PV001205 Actual Droop
1	4	487	PV001137 T-Coolant Intercooler	PV001169 T-ECU

Table 8: Node ID Offset 1

Node ID Offset 2

Node ID Offset	Transmit PDO	COB ID	PV Name	PV Name
2	1	188	PV001012 Speed Demand Source	PV001075 Actual Failure Codes
2	2	288	PV001219 Nominal Power	PV001409 Requested Torque
2	3	388	PV001195 Feedback Speed Demand ana.CAN PV001453 Engine Power *****	PV001196 Feedback Speed Demand analog PV001120 Daily Fuel Consumption *****
2	4	488	PV001063 ECU Operating hours (sec)	PV001104 ECU Power Supply Voltage

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Table 9: Node ID Offset 2

Node ID Offset 3

Node ID Offset	Transmit PDO	COB ID	PV Name	PV Name
3	1	189	PV005352 Level Day-Tank %	PV005362 Level Holding Tank %
3	2	289	PV006035 Pressure 1 PV006035 T-Bearing Geno DE *****	PV006055 Pressure 2 PV006055 T-Bearing Geno NDE *****
3	3	389	PV006010 AIN T-Exhaust A	PV006020 AIN T-Exhaust B

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